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Tech Overview

Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

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Summary

The paper proposes Rep-YOLO11s, a real-time helmet detector for construction surveillance that integrates four known components: structural re-parameterization (RepConv) to fuse multi-branch training blocks into single 3×3 convolutions at inference, CoordConv to inject positional priors, BiFormer bi-level routing attention to emphasize salient tokens, and Focal EloU for bounding-box regression. On SHWD, the authors report mAP50 up to 94.83% (single run) and 96.64% (5-fold mean), strong throughput on GPUs (2.92–5.35 ms FP16 TensorRT at 640×640), and zero-shot transfer to multiple external datasets. They also provide detailed deployment measurements and an end-to-end RTSP pipeline analysis.

Strengths

Technical novelty and innovation

- The paper combines structural re-parameterization, CoordConv, and BiFormer within a YOLO11s-style framework aimed at small, distant objects in cluttered industrial scenes.
- Clear algebraic exposition of RepConv fusion steps (Conv+BN folding, 1×1 padding, identity Dirac kernel) that is implementation-ready.
- Practical attention to deployment (TensorRT FP16, CUDA synchronization) and pipeline bottlenecks is valuable for industrial informatics.

Experimental rigor and validation

- Broad set of evaluations: single-run and 5-fold cross-validation on SHWD, plus zero-shot tests across five external datasets.
- Multi-platform benchmarking (T4, RTX 3050 Laptop GPU, CPU) with synchronized timing and per-stage pipeline latency breakdown is commendable.
- Ablation study touches each added component (CoordConv, RepConv, Focal EloU, BiFormer).

Clarity of presentation

- The architecture is explained step-by-step with equations and system-level diagrams; the deployment pipeline stages and timing are described in detail.
- The discussion of synchronization pitfalls for GPU timing is practical and informative.

Significance of contributions

- Addresses a relevant industrial safety problem where real-time performance and robustness to small objects/occlusions are critical.
- The deployment-focused evaluation (including end-to-end RTSP pipeline) is aligned with IEEE TII's emphasis on practical applicability.

Weaknesses

Technical limitations or concerns

- Conceptual novelty is limited: all core components (RepConv, CoordConv, BiFormer, Focal EloU) are established; the contribution is primarily an integration/tailoring to YOLO11s.
- The claim that Focal EloU resolves severe class imbalance is conceptually incorrect; it is a regression loss and does not address class-imbalance in classification directly.
- The impact of BiFormer at inference is under-quantified; attention modules add compute and memory access, and cannot be “re-parameterized away” like RepConv.

Experimental gaps or methodological issues

- Metric inconsistencies across tables are concerning (e.g., mAP50–95 on SHWD reported as 62.54% in Table I vs 77.90% in Table III without clear protocol differences), calling into question result comparability.
- SOTA comparisons mix measurement conditions; it is unclear if all latencies were measured under identical engines/hardware and if models were re-trained under matched protocols.
- EC-YOLOv8 is reported with much higher mAP50–95 (74.6%) than the proposed method (62.54%), yet the narrative claims superiority “in both accuracy and throughput,” which is not supported by the table.
- The ablation gains are small and sometimes within noise; no variance/confidence intervals are reported for ablations.

Clarity or presentation issues

- Multiple typographical inconsistencies (YOLO1ls vs YOLO11s; “EloU” vs EloU) and occasional contradictory statements (e.g., saliency analysis on posters) reduce clarity.
- Lack of precise BiFormer hyperparameters (region size S , top- k) and insertion points; unclear where CoordConv is injected and at what cost across layers.

Missing related work or comparisons

- Important related/backbone work like YOLOv6/EfficientRep (rep-friendly backbones and necks for TensorRT/T4) and recent helmet detectors (e.g., DABFNet) or reparameterized necks (e.g., MAF-YOLO) are not compared experimentally; this diminishes the context for claimed speed/accuracy trade-offs.

- No evaluation on prevalent embedded edge platforms for industrial deployment (e.g., NVIDIA Jetson series), which are highly relevant to the paper's stated use case.

Detailed Comments

Technical soundness evaluation

- The RepConv derivation is standard and correct; merging Conv-BN, padding 1×1 to 3×3 , and adding an identity Dirac kernel are well-described. CoordConv augmentation is consistent with prior art and relevant to spatial priors in PPE detection. Focal EloU is correctly formulated for regression; however, the manuscript over-attributes class-imbalance benefits to it. BiFormer's bi-level routing is briefly but correctly summarized; nonetheless, the practical cost at inference on GPU/edge is not isolated.
- The integration choices (where in the neck to place attention; which layers receive CoordConv) should be specified precisely; these details matter for both performance and latency.

Experimental evaluation assessment

- The breadth of experiments is a strength, especially the deployment profiling and pipeline-level latency accounting with synchronized CUDA timing. However, there are internal inconsistencies:
 - SHWD mAP50–95 varies drastically across tables without transparent protocol changes. This must be reconciled (e.g., different class sets, person-vs-hat evaluation, altered IoU thresholds, different test splits).
 - Claims of being best in “both accuracy and throughput” are contradicted by Table I (e.g., EC-YOLOv8's higher mAP50–95).
 - Ablation improvements are marginal (≈ 0.1 – 0.2 points), and latency reporting mixes pre-deploy and post-deploy settings across variants; per-component inference-time costs on a single, consistent engine are needed.
- Cross-domain results are appreciated and realistic (substantial variation across datasets). The discussion on annotation protocol misalignment is

apt, but the exact harmonization rules and evaluation scripts should be provided for reproducibility.

Comparison with related work (using the summaries provided)

- RepVGG/RepConv (2101.03697, YOLOv6/EfficientRep 2302.00386) show that rep-style single-path inference is excellent for real-time; a fair comparison against YOLOv6-N/S on Tesla T4/TensorRT would strengthen the case for Rep-YOLO11s.
- BiFormer (2303.08810) is a reasonable choice for dynamic sparse attention; nevertheless, its practical latency overhead should be quantified relative to alternative light attention (e.g., ECA) and compared against DAHead/BWFPN (DABFNet, 2411.19071).
- MAF-YOLO (2407.04381) and other re-parameterized small-object focused necks are relevant comparisons for small/distant targets; an empirical comparison would contextualize the claimed advantages.

Discussion of broader impact and significance

- The paper's industrial value proposition is strong: real-time, cross-domain robustness, and careful deployment benchmarking. The integrity discussion about synchronized GPU timing addresses a common pitfall in the community.
- To fully meet TII standards, the study should validate on representative embedded edge devices (e.g., Jetson Orin/NX/Xavier) with power/throughput trade-offs and, ideally, multi-camera scaling behavior in realistic conditions.

Questions for Authors

1. Please reconcile the discrepancies in SHWD mAP50–95 across tables (e.g., 62.54% vs 68.26% vs 77.90%). What exact evaluation protocols, class sets, and IoU thresholds were used in each table?
2. Were all baseline latencies measured with the same engine (TensorRT version/precision), batch size, input resolution, and GPU? If not, please re-run or clearly annotate to ensure a fair apples-to-apples comparison.

3. What are the exact BiFormer hyperparameters (region size S , top- k), insertion points, and channel dimensions? Please report its isolated inference-time overhead on TensorRT/PyTorch and its ablation impact on both accuracy and latency.
4. You state Focal EloU addresses severe class imbalance. Since it is a regression loss, how does it mitigate classification imbalance? Did you consider focal classification loss or reweighting, and what were the effects?
5. In Table II, "Full Fusion" shows a large latency drop after switch_to_deploy and TensorRT FP16. Could you provide per-component latencies measured with the same engine (e.g., TensorRT FP16 for all ablations) to cleanly attribute runtime costs?
6. Please detail the 5-fold cross-validation setup: exact splits, class distribution per fold, and whether any hyperparameter/model selection used test folds (to avoid leakage).
7. For cross-domain evaluations, what precise label harmonization steps were taken (e.g., person vs head vs hat)? Can you share the evaluation scripts and mapping rules to enable reproduction?
8. The Grad-CAM discussion for signage/posters seems contradictory (detecting helmets on posters). How often are such FPs observed, and can you incorporate person-helmet spatial association constraints to suppress them?
9. Could you add comparisons against YOLOv6/EfficientRep and recent helmet detectors (e.g., DABFNet, MAF-YOLO) on SHWD using identical training and evaluation settings?
10. Please include results on Jetson-class devices (e.g., Orin NX/AGX) with measured power and batch=1 throughput to demonstrate real-world edge feasibility.
11. Where specifically are CoordConv channels injected (which layers), and what is the incremental latency and memory overhead? Any ablation on that placement?
12. Will you release code, trained weights, TensorRT engines, and the synchronized timing scripts to support reproducibility?

Overall Assessment

This manuscript targets a timely and important industrial application and demonstrates commendable attention to real-world deployment details, including synchronized GPU timing and end-to-end pipeline profiling. The integration of rep-style blocks, CoordConv, and BiFormer is reasonable for small-object detection in cluttered scenes, and the extensive cross-domain evaluation is a welcome attempt to assess robustness. However, the work's methodological novelty is modest, and several critical issues limit its suitability in its current form: internal inconsistencies in reported metrics, potentially unfair SOTA comparisons, over-claims (e.g., about class-imbalance mitigation), and incomplete quantification of attention overhead at inference. The ablation gains are small and lack statistical support, and key related baselines (e.g., YOLOv6/EfficientRep, recent helmet detectors) are not compared under a unified protocol. With a thorough cleanup of evaluation inconsistencies, standardized and fair baseline comparisons, precise reporting of attention hyperparameters/overhead, and added edge-device validation, this work could provide solid practical value. As submitted, I recommend a major revision.

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